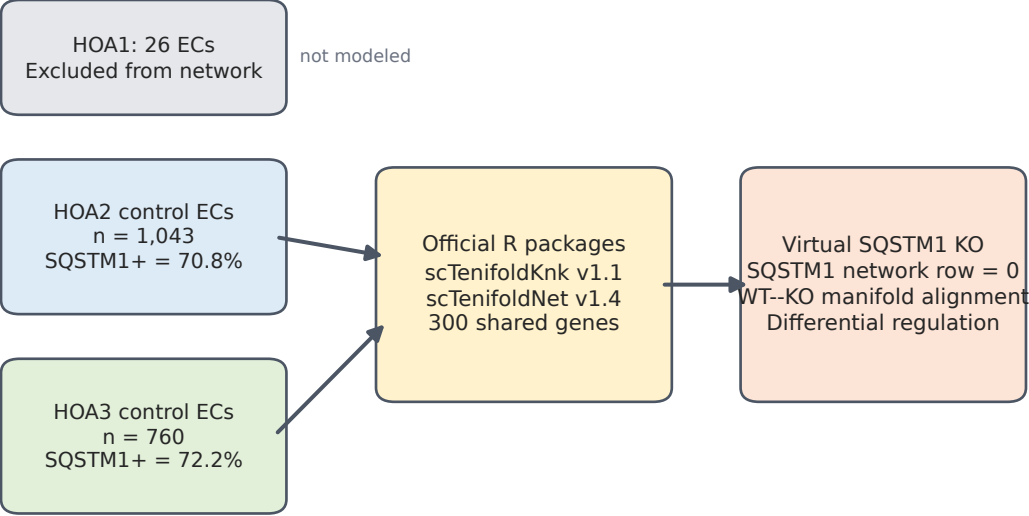


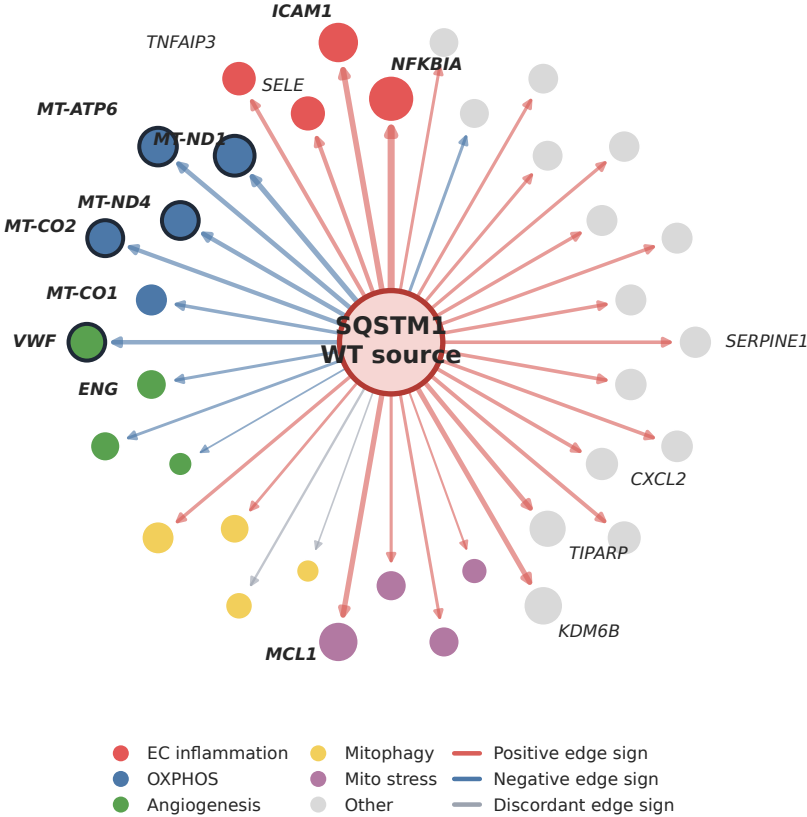
A Official-package virtual-knockout design



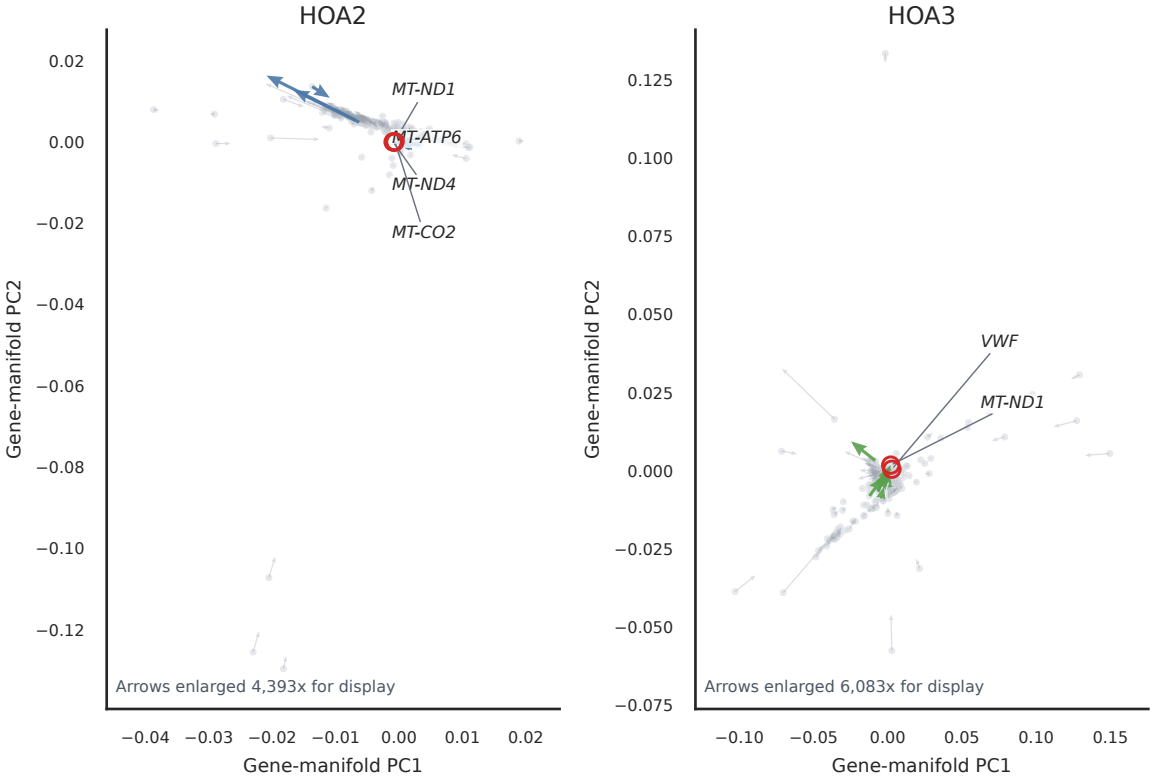
Parameter audit

Primary: 20 networks, $q=0.95$, 30 manifold dimensions | Package-default sensitivity: 10 networks, $q=0.90$, 2 dimensions
No downstream gene or pathway was FDR-significant in both donors under both profiles.

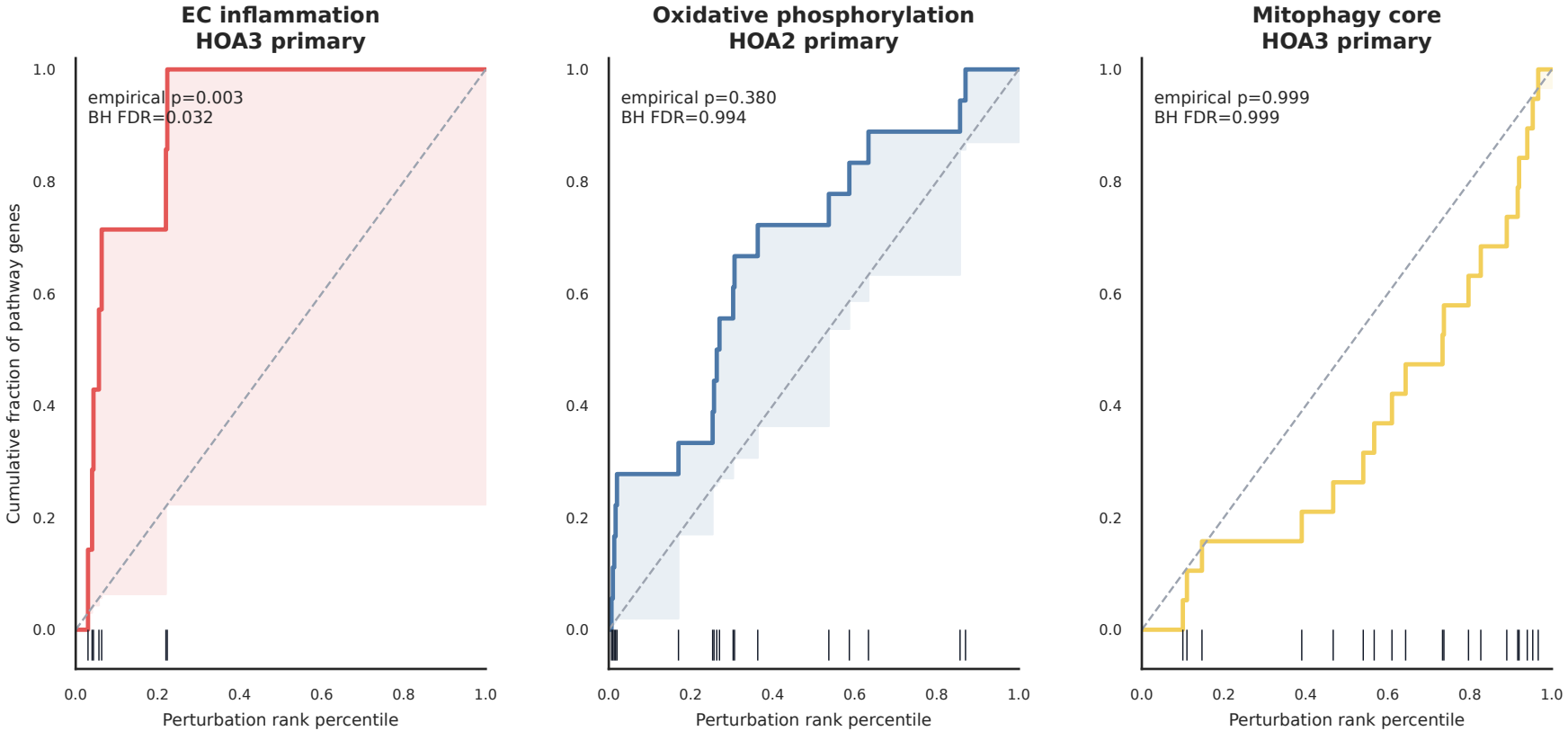
B SQSTM1 WT neighborhood removed by virtual KO



C WT-to-virtual-KO displacement in the aligned gene manifold



D Matched-null pathway rank recovery



Gray arrows: downstream genes; colored arrows: top 10 ranks; red rings/labels: $FDR<0.05$.
Uniform magnification preserves direction/relative size; these are gene-manifold displacements, not cell-fate vectors.

Curves show pathway-member recovery along official perturbation ranks; p /FDR use 20,000 matched null sets.
These descriptive curves are not GSEA enrichment-score curves.